

Lab 13

Multiple Regression Analyses

Walkthrough (5 points)

Instructions

Walk Through → Data come from the **huskers** dataset, including all games after Nebraska's last national championship season (post-1997).

Important: Before running any analyses, you must select only games after the 1997 season. In SPSS, use **Data** → **Select Cases** → **If condition is satisfied**, and enter the condition `season > 1997`. Click **Continue**, then **OK**. You should see N = 340 cases remaining.

Criterion variable is **Nebraska Total Score**

Predictor variables:

- **Nebraska rushing yards** (`ne_rush_yards`)
- **Nebraska passing yards** (`ne_pass_yards`)
- **Temperature at kickoff (Fahrenheit)** (`temp`)
- **Conference game** (`conference`) — Values: 0 = Non-conference, 1 = Conference game
- **Whether it was a home game** (`home`) — Values: 0 = No, 1 = Yes

Step 1: Univariate Analysis

Get the frequencies for each variable. In the table below indicate whether each variable is quantitative or binary.

Step 2: Bivariate Analyses

Get the correlations (and p-values) between the criterion and each predictor. Put those values in the table below.

Step 3: Multivariate Analysis

Perform the multiple regression analysis and fill in the rest of the table below.

Table 1

Predictor	Is variable quant or binary?	r (p) With Nebraska Total Score	b (p)	Bivariate/Multivariate Results Categories: a. Neither r nor b significant b. r & b both sig & same sign c. r sig but not b d. suppressor effect
Nebraska rushing yards				
Nebraska passing yards				
Temperature at kickoff (Fahrenheit)				
Conference game (non-conference = 0, conference = 1)				
Home game (no = 0, yes = 1)				

R = _____ R² = _____ F = _____ df = _____, _____ p = _____

a. Does the multiple regression model work? Where did you look to decide?

Answer:

b. How well does the multiple regression model work?

Answer:

c. Interpret each of the correlations (use language appropriate to the binary or quant predictor being interpreted)

Table 2

Predictor	Interpretation
Nebraska rushing yards	
Nebraska passing yards	
Temperature at kickoff (Fahrenheit)	
Conference game (non-conference = 0, conference = 1)	
Home game (no = 0, yes = 1)	

d. Interpret each of the regression weights:

Table 3

Predictor	Interpretation
Nebraska rushing yards	
Nebraska passing yards	
Temperature at kickoff (Fahrenheit)	
Conference game (non-conference = 0, conference = 1)	
Home game (no = 0, yes = 1)	

Your Turn (10 points)

Instructions

Your Turn → Data come from the **huskers** dataset, including all games after Nebraska's last national championship season (post-1997).

Important: Make sure you still have the post-1997 filter active (Select Cases: season > 1997). You should have N = 340 cases.

Criterion variable is **Opponent Total Score**

IVs:

- **Opponent rushing yards** (opp_rush_yards)
- **Opponent passing yards** (opp_pass_yards)
- **Nebraska penalty yards** (ne_pen_yards)
- **Temperature at kickoff (Fahrenheit)** (temp)
- **Whether Nebraska won** (win) — Values: 0 = No, 1 = Yes
- **Whether it was a home game** (home) — Values: 0 = No, 1 = Yes

Univariate Analysis

Get the frequencies for each variable. In the table below indicate whether each variable is quantitative or binary.

Bivariate Analyses

Get the correlations (and p-values) between the criterion and each predictor. Put those values in the table below.

Multivariate Analysis

Perform the multiple regression analysis and fill in the rest of the table below.

Table 4

Predictor	Is variable quant or binary?	r (p) With Opponent Total Score	b (p)	Bivariate/Multivariate Results Categories: a. Neither r nor b significant b. r & b both sig & same sign c. r sig but not b d. suppressor effect
Opponent rushing yards				
Opponent passing yards				
Nebraska penalty yards				
Temperature at kickoff (Fahrenheit)				
Nebraska won (no = 0, yes = 1)				
Home game (no = 0, yes = 1)				

R = _____ R² = _____ F = _____ df = _____, _____ p = _____

a. Does the multiple regression model work? Where did you look to decide?

Answer:

b. How well does the multiple regression model work?

Answer:

c. Interpret each of the correlations (use language appropriate to the binary or quant predictor being interpreted)**Table 5**

Predictor	Interpretation
Opponent rushing yards	
Opponent passing yards	
Nebraska penalty yards	
Temperature at kickoff (Fahrenheit)	
Nebraska won (no = 0, yes = 1)	
Home game (no = 0, yes = 1)	

d. Interpret each of the regression weights:**Table 6**

Predictor	Interpretation
Opponent rushing yards	
Opponent passing yards	
Nebraska penalty yards	
Temperature at kickoff (Fahrenheit)	
Nebraska won (no = 0, yes = 1)	
Home game (no = 0, yes = 1)	

Grab screenshots of the homework checker screens, showing all of your input answers in green (correct). Paste them all here. Your TA will not grade the assignment without this.

Paste Homework Checker screenshots here: